

# Verification Test Protocol 2

## *Class D Amplifier Pre-Power and Dead Time Verification*

|              |                                                       |
|--------------|-------------------------------------------------------|
| Project      | EE22005 Home Audio System (HAS) — Project Week 4      |
| Group        | Group HAS (BTA26, FS881, AC3847)                      |
| Document ID  | 20260430_HAS_ClassD_TestProtocol_v1.docx              |
| Version      | v1.0                                                  |
| Prepared for | Feasibility Study — Synthesis Evidence (Verification) |

| Version | Date       | Author                           | Description of changes                                                                                            |
|---------|------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| v1.0    | 30/04/2026 | Group HAS (BTA26, FS881, AC3847) | Initial issue of Class D pre-power continuity, triangle-wave, PWM, dead-time and DC-offset verification protocol. |

|                       |                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test ID               | PW4_HAS_VER_002                                                                                                                                                                                                      |
| Document purpose      | Formal pre-power verification of the Class D amplifier wiring and powered verification of triangle-wave generation, PWM modulation, dead-time correctness, and DC offset at the load before any earphone connection. |
| Related specification | S2 — Class D switching efficiency and PWM correctness; S3 — RLC output filter; S6 — Hearing-safety output limits.                                                                                                    |
| Verification type     | Combined pre-power continuity check and powered oscilloscope verification with hold-point before earphone connection.                                                                                                |
| Primary equipment     | Digital multimeter, four-channel oscilloscope, function generator (1 kHz sine source), bench supply, oscilloscope cursors, no earphones connected during this protocol.                                              |
| Prepared by           | Group HAS (BTA26 — build lead; FS881 — bench                                                                                                                                                                         |

|                    |                                                     |
|--------------------|-----------------------------------------------------|
|                    | verification lead; AC3847 — evidence custodian)     |
| Revision / Version | v1.0 — Initial issue, end of Day 3 / start of Day 4 |

## 1. Objective

To verify that the Class D switching amplifier is electrically correct before power is applied, that the triangle-wave generator and PWM comparator operate to specification, that the half-bridge gate-drive includes sufficient dead time to prevent shoot-through, and that the DC offset at the RLC filter output is within the safe limit required for earphone connection. Earphones shall not be connected at any point during this protocol [1], [3], [4], [9].

## 2. Requirement traceability

| Ref | Requirement summary                        | Acceptance criterion                                             | Compliance basis                                 |
|-----|--------------------------------------------|------------------------------------------------------------------|--------------------------------------------------|
| S2  | Continuity on intended Class D nets        | < 1 $\Omega$ before power is applied                             | DMM continuity check on each critical node.      |
| S2  | Insulation between supply rails and ground | > 1 M $\Omega$ before power is applied                           | DMM resistance check, system unpowered.          |
| S2  | Triangle wave at integrator output         | 100 kHz $\pm 5$ %, peak-to-peak within $\pm 10$ % of design      | Oscilloscope cursor measurement.                 |
| S2  | PWM correctly modulated by 1 kHz test sine | Visible duty-cycle modulation tracking the input                 | Two-channel oscilloscope capture (input vs PWM). |
| S2  | Dead time on both transitions              | $\geq 0.5$ $\mu$ s on rising and falling edges, no shoot-through | Oscilloscope cursor measurement on both gates.   |
| S6  | DC offset at RLC filter output             | < 0.4 V before earphones are connected                           | DC-coupled oscilloscope reading; hold-point.     |

## 3. Roles and responsibilities

Test lead (FS881): owns hold-point sign-off and final DC offset confirmation. Build lead (BTA26): operates equipment and performs any rework on the half-bridge. Evidence custodian (AC3847): captures oscilloscope screenshots of triangle wave, PWM, dead-time, and DC offset, and saves them to the QMS Quality\_Control folder per the naming convention [10]. The earphone-connection hold-point may only be released by the test lead.

## 4. Equipment and preconditions

System fully unpowered for Sections 6.1–6.3. Bench supply set to off, USB removed.

Latest Class D schematic available at the bench.

Digital multimeter verified operational and set to continuity / resistance ranges as required.

No earphones present on the bench during this protocol.

Oscilloscope ground reference confirmed common with the Class D analogue ground.

## **5. Controlled variables**

System remains unpowered throughout Sections 6.1 to 6.3 (pre-power phase).

Same multimeter retained for the full pre-power phase.

Probe pressure kept light and consistent to avoid false continuity readings.

Function-generator amplitude held at 100 mV pp / 1 kHz throughout the PWM verification.

No rewiring permitted during measurement except where rework is documented.

## **6. Detailed procedure**

### **6.1 Visual inspection (unpowered)**

Inspect the breadboard against the Class D schematic. Confirm no shorts between adjacent rails, no reversed-polarity electrolytic capacitors, no exposed wire bridging the half-bridge nodes, and that the gate-drive RC dead-time network values match the schematic.

### **6.2 Continuity on intended nets (unpowered)**

Set the DMM to continuity. Verify continuity ( $< 1 \Omega$ ) on the following nets: +V supply rail, –V supply rail, analogue ground return, integrator output to comparator input, comparator output to gate driver, half-bridge midpoint to RLC filter input, RLC filter output to DC blocking capacitor input. Record each measurement in Section 7.

### **6.3 Rail-to-ground insulation (unpowered)**

Switch DMM to resistance. Confirm that resistance from each supply rail to ground exceeds  $1 \text{ M}\Omega$  with the system unpowered. Both +V  $\rightarrow$  ground and –V  $\rightarrow$  ground shall pass before power is applied [1].

### **6.4 Triangle-wave verification (powered)**

Apply bench supply rails. Probe the integrator output with the oscilloscope. Measure period and amplitude using cursors. Confirm 100 kHz  $\pm 5 \%$  and amplitude within  $\pm 10 \%$  of the design target. Capture screenshot M3a per [10].

### **6.5 PWM modulation verification (powered)**

Inject a 1 kHz, 100 mV pp sine into the comparator input. Probe the comparator output. Confirm a square wave whose duty cycle is visibly modulated by the input sine. Capture a two-channel screenshot showing both signals.

### **6.6 Dead-time verification (powered)**

Probe the high-side gate-drive node on Channel 1 and the low-side gate-drive node on Channel 2. Trigger on either channel's rising edge. Use cursors to measure the interval during which both gate signals are simultaneously below their conduction thresholds. Confirm dead time  $\geq 0.5 \mu\text{s}$  on both rising and falling transitions, with no overlap. If overlap is observed, halt and rework the dead-time RC values before re-running this section [3], [4]. Capture screenshot M3b per [10].

### 6.7 DC offset at RLC filter output (powered, hold-point)

With the RLC filter built and the load resistor (representing earphone impedance) in place but no earphones connected, measure the DC voltage at the RLC filter output node using the oscilloscope DC-coupled. Confirm offset  $< 0.4 \text{ V}$  under no-signal, low-signal (100 mV input) and full-signal conditions. This is a hold-point: no earphones may be connected until all three readings are confirmed below 0.4 V and signed off by the test lead [1], [9]. Capture screenshot M4a per [10].

### 6.8 Rework and re-verification

If any value fails its acceptance criterion, isolate the affected stage, document the suspected fault location, perform rework, and repeat the affected sections in full. If a power-rail or DC-offset fault is identified, re-run the full protocol from Section 6.1.

## 7. Data recording template

| Check item                        | Expected criterion     | Measured value         | Pass / Fail | Notes / rework |
|-----------------------------------|------------------------|------------------------|-------------|----------------|
| +V rail continuity                | $< 1 \Omega$           | _____ $\Omega$         | _____       |                |
| –V rail continuity                | $< 1 \Omega$           | _____ $\Omega$         | _____       |                |
| Analogue ground continuity        | $< 1 \Omega$           | _____ $\Omega$         | _____       |                |
| Half-bridge midpoint to RLC input | $< 1 \Omega$           | _____ $\Omega$         | _____       |                |
| +V rail to ground (insulation)    | $> 1 \text{ M}\Omega$  | _____ $\text{M}\Omega$ | _____       |                |
| –V rail to ground (insulation)    | $> 1 \text{ M}\Omega$  | _____ $\text{M}\Omega$ | _____       |                |
| Triangle wave frequency           | 100 kHz $\pm 5 \%$     | _____ kHz              | _____       |                |
| Triangle wave amplitude           | Per design $\pm 10 \%$ | _____ V pp             | _____       |                |
| PWM duty cycle modulation visible | Yes (qualitative)      | _____                  | _____       |                |
| Dead time — rising transition     | $\geq 0.5 \mu\text{s}$ | _____ $\mu\text{s}$    | _____       |                |

|                                 |                        |                     |       |  |
|---------------------------------|------------------------|---------------------|-------|--|
| Dead time — falling transition  | $\geq 0.5 \mu\text{s}$ | _____ $\mu\text{s}$ | _____ |  |
| Shoot-through current observed  | None                   | _____               | _____ |  |
| DC offset — no signal           | $< 0.4 \text{ V}$      | _____ $\text{V}$    | _____ |  |
| DC offset — low signal (100 mV) | $< 0.4 \text{ V}$      | _____ $\text{V}$    | _____ |  |
| DC offset — full signal         | $< 0.4 \text{ V}$      | _____ $\text{V}$    | _____ |  |

## 8. Acceptance criteria and decision rule

All continuity measurements shall be below  $1 \Omega$ .

All rail-to-ground insulation measurements shall exceed  $1 \text{ M}\Omega$  when unpowered.

Triangle-wave frequency shall be within  $100 \text{ kHz} \pm 5 \%$ .

PWM duty cycle shall visibly track the test sine.

Dead time shall be  $\geq 0.5 \mu\text{s}$  on both transitions with no shoot-through.

DC offset at the RLC output shall remain below  $0.4 \text{ V}$  across all three signal conditions.

Any failed item requires correction and re-verification before earphones may be connected.

## 9. Safety notes

Do not connect earphones at any point during this protocol [1], [9].

Do not probe a circuit that has become accidentally powered during the unpowered phase; isolate first.

Avoid slipping probes across adjacent half-bridge pins — risk of momentary short across the supply.

Half-bridge transistors may become warm under sustained operation; check temperature with caution at end of session.

## 10. Conclusion statement

The Class D amplifier is considered fit for connection to the earphone load only when every check in this protocol — continuity, insulation, triangle wave, PWM modulation, dead time, and DC offset — has been completed, recorded, and passed without unresolved anomalies. The earphone-connection hold-point may be released only by the test lead after all three DC-offset readings are signed off below  $0.4 \text{ V}$ .

## References

- [1] International Electrotechnical Commission, Audio/video, information and communication technology equipment — Part 1: Safety requirements, IEC 62368-1:2023, 4th ed., May 2023.

- [3] D. Self, Audio Power Amplifier Design, 6th ed. Oxford, U.K.: Focal Press, 2013, ch. 12.
- [4] Texas Instruments, "Class-D LC filter design," Application Report SLOA119B, Apr. 2017.
- [9] World Health Organization, "Make Listening Safe," Geneva, Switzerland, 2015.
- [10] Department of Electronic and Electrical Engineering, EE22005 Engineering Practice and Design — Project Week 4 Brief, Univ. of Bath, 2026.